

Scaling AI Beyond the Data Center

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Why Multi-Site AI Breaks Reactive Networks

The Scale-Across reality: power and space caps force AI clusters beyond a single campus

The Latency Tax

10km Fiber adds 100s of µs RTT

PFC/ECN feedback loops are too slow to react before queues overflow at that RTT.

Fan-In Bursts

AI collectives (AllReduce/AllGather) create synchronized fan-in spikes.

Shallow on-chip buffers exhaust in microseconds under burst.

Failures Are the Norm, Not the Exception

At this scale, link/path failures happen continuously.

Even the best proactive routing must still REACT to a failure.

At 10km+ RTT, that reaction window is where headroom is needed.

Why Hierarchical Buffering, Not All-HBM

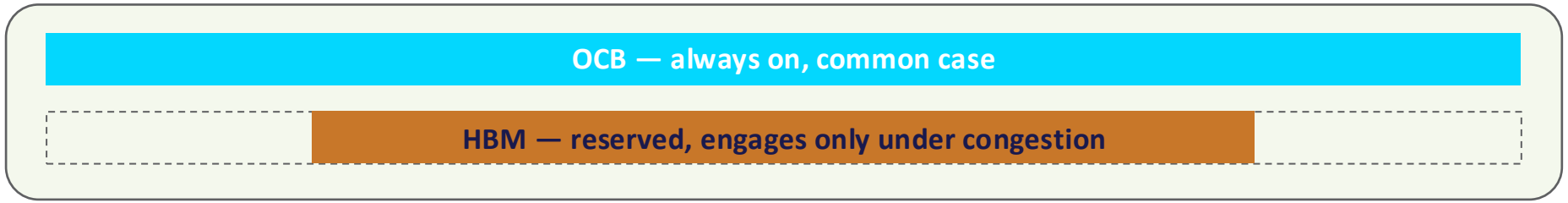
Headroom matters for AI collectives — but putting it on every port doesn't.

① Why Headroom Matters

- One dropped packet stalls the whole job — but most traffic never gets close to congestion.

② Why Not All-HBM

- HBM per port = expensive, power-hungry, hard to route.
- OCB handles the common case; HBM stays in reserve.



Hierarchical, Hybrid Buffering

Shallow Speed, Deep Insurance

On-chip buffer for the common case; HBM engaged from 10km onward, where distance and failures demand

In Production Today (Hierarchical by Design)

Intra-building:

shallow buffer, no oversubscription.

Building-to-building (<10km):

shallow buffer, light oversubscription, telemetry-managed.

Region-to-region (10km+):

HBM deep buffer, multiple RTT headroom, any oversub.

Buffer Math (10km Regional Link)

Link speed: 800 Gb/s

RTT (10km): 100 µs

Required buffer / port (BDP): 10 MB

Actual buffer / port (fixed HW): 200 MB

HBM capacity available: up to 8.0 GB



Deep Buffers Minimize Job Completion Time

JCT is set by the slowest path, not the average – and drops are far more expensive than queuing delay

Why This Is True (Mechanism)

JCT = time of the SLOWEST path, not the mean

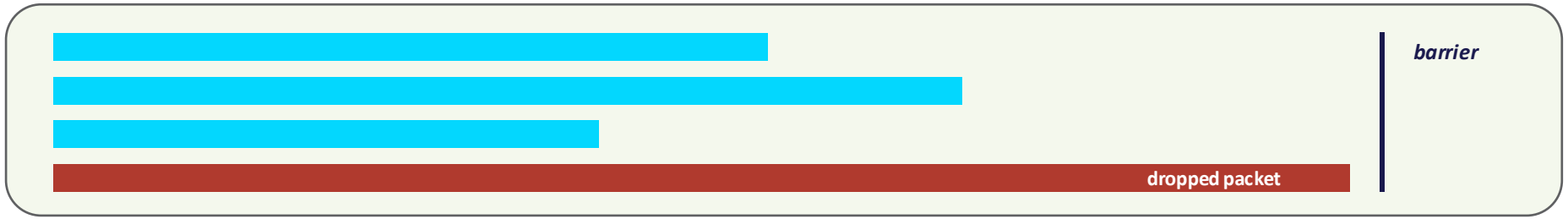
- Jitter isn't the dominant driver at this scale.

A drop stalls the whole collective until retransmission

- a queued burst costs microseconds
- a drop costs **orders of magnitude** more.

Proven in Simulation

...details shortly...



The Front End Network Feeds the AI Fabric

Every AI fabric sits behind a front-end network stitching together storage, compute, and the outside world

What Feeds Training

Local storage: for checkpointing and training data

General-purpose compute: for preprocessing training data.

Cloud storage: (S3, Azure, GCP) for raw, unprocessed data.

Corporate Network: for fine tuning

What Serves Inference

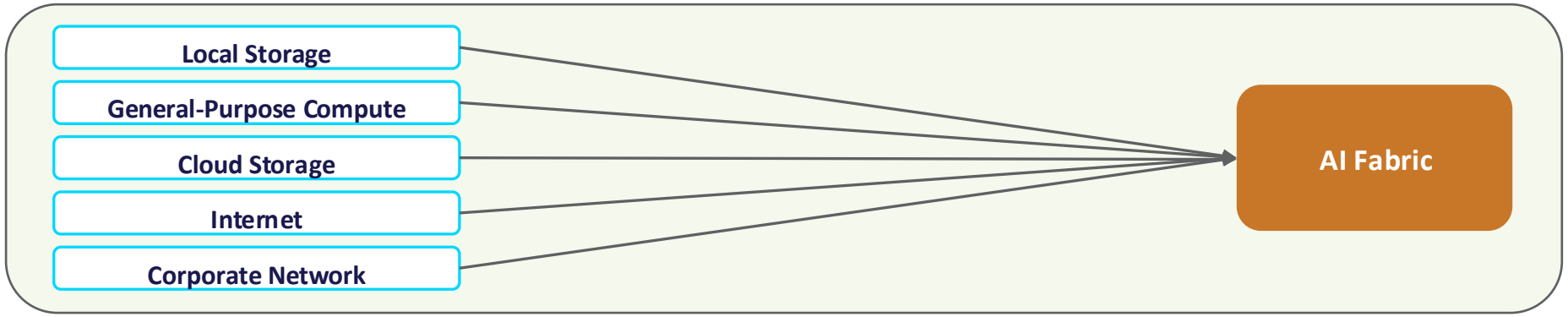
Internet: real-time queries and results.

Secure corporate network: private data for inference, training, fine-tuning, RAGs, and streaming inference.

Compute: for Agents and Tools

Cloud storage: for RAGs, user attachments

Local Storage: KV cache offload



Deep Buffers for the Front End

A heterogeneous mix of protocols and distances - not the single tuned RDMA fabric this talk has covered so far

A Different Kind of Traffic

Multiple protocols (TCP, RDMA, NVMe, UEC, IPSec)

Mix of traffic on links

Not one single tuned RDMA fabric

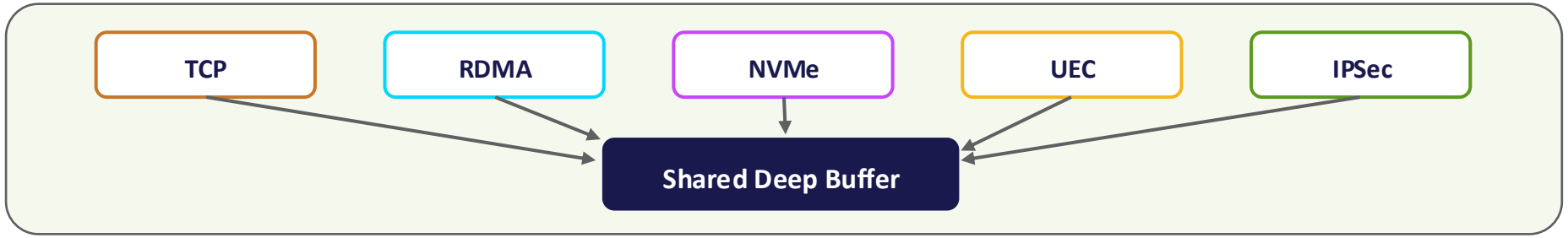
Where Buffers Earn Their Keep

Mostly

- single-flow
- uneven distances
- less-controlled congestion

deep buffers are insurance, for a messier mix of traffic

often at the edge, where front-end meets Metro/WAN



Simulation Setup: Buffer-Size Sweep

2:1 long-haul oversubscription, buffer size sweep in units of BDP — tested at 10km, 30km, and 100km

Topology & Traffic

Traffic over the long-haul only — no competing local traffic.

Packet spraying with Advanced CC (NSCC and RCCC)

Buffer size sweep from 3×BDP down to 0.005×BDP at each distance.

Repeated independently at 10km, 30km, and 100km

Additional sweep of message size – larger and smaller relative to BDP

What's Being Measured

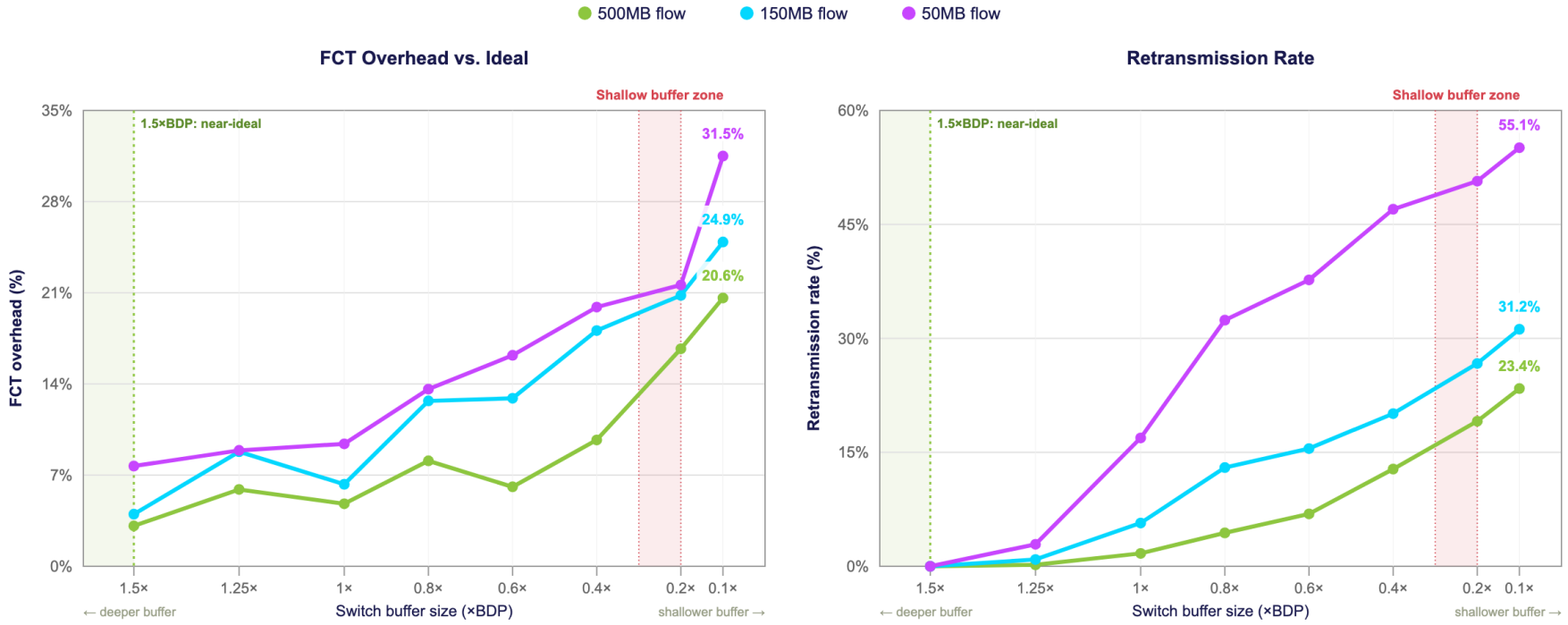
Flow Completion Time & Retransmission rate

Buffer requirement relative to BDP with advanced CC

Performance at smaller message sizes relative to BDP

Buffer Depth Determines Completion Time

Buffer ≥1.5×BDP keeps FCT near-ideal and retransmission at zero on all sweeps
When buffer is smaller - Significant increase in FCT and Retransmission rate – especially in smaller messages



Message-size sweep run at 10km under NSCC. The ≥1.5×BDP threshold itself is confirmed at 10km, 30km, and 100km, under both NSCC and RCCC.

Key Takeaway: Headroom Beats Retransmission

The simulation isolates the minimum buffer required — Consistent over distances and with both advanced CC protocols

① Near-Ideal FCT
Buffer ≥1.5×BDP lands within a few percent of ideal — at every distance tested.

② Retransmission First
Below threshold, retransmission climbs before completion time does — then both grow sharply.

③ CC-Agnostic
The threshold holds under NSCC and RCCC alike — a buffer-sizing property, not a tuning trick.

Buffer necessary @ 100Tb switch (1.5xBDP)

10km	30km	100km
~1.9GB	~5.6GB	~18.8GB

Shallow Buffer switches can't support the buffer necessary at these distances

A System, Not a Switch

Orchestration, oversubscription, and utilization – solved end-to-end, not one piece at a time

Thor™ NIC

RCCC+ rate modulation across long RTT.

Speculative transmission to hide handshake latency.

HW tracking of in-flight packets for wire-speed retransmission.

Interconnect – Hierarchical Switch

Shallow or hierarchical buffering — whichever the deployment calls for, not one fixed answer.

HyperPort™ consolidation improves hashing/ECMP entropy for elephant flows.

Trims and re-injects bursts instead of relying on PFC pause frames.

Back to Sender – expedite feedback

Unified Stack

Telemetry-driven modulation (INT/CSIG) from real-time queue depth.

Integrates with customer orchestration layers.

Thor and HyperPort are trademarks of Broadcom Inc.

The Foundation for the Path to AGI

A joint industry framework, not a single-vendor pitch

The System-Level Challenge

Regional AI scaling cannot be solved by switch buffer depth alone, nor by NIC-side telemetry alone – it requires both, coordinated.

The Hierarchical, Hybrid Buffer Architecture

Shallow on-chip buffers for the common case; HBM deep buffers as tier-appropriate insurance at the region-to-region boundary where RTT and failure risk are highest. Broadcom and Arista jointly validate this model in production.

Built for 1M+ GPU Regional Clusters

A scalable architecture – fixed box to chassis to distributed fabric – engineered for the transition to gigawatt-scale, multi-site AI infrastructure.

Get Involved & Learn More

- **Call to Action: Bring us your scale-across topology — we'll help size buffer depth and oversubscription for it.**
- Broadcom - What Is Scale-Across Networking for AI Clusters: https://www.broadcom.com/topics/what-is-scale-across-networking-for-ai-clusters
- Arista - The Many Facets of AI Fabrics: https://blogs.arista.com/blog/the-many-facets-of-ai-fabrics
- Questions? Find Golan (Broadcom) or Hugh (Arista) after this session - we'd love to talk through your deployment.

Thank You!



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